CG	CLASS INDEX NO.
07S ___	

CANDIDATE NAME: ______________________________

TAMPINES JUNIOR COLLEGE
PRELIMINARY EXAMINATION 2008

BIOLOGY 9747/02
Paper 2 Core Paper

THURSDAY 21 AUGUST 2008 **11.30 AM – 1.30 PM (2 hours)**

Additional materials: Answer paper

INSTRUCTIONS TO CANDIDATES

(a) Write your name, Civics group and candidate number in the spaces at the top of this page and on all separate answer paper used.
(b) Write in dark blue or black pen on both sides of the paper.
(c) You may use a soft pencil for any diagrams, graphs or rough working.
(d) Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A
Answer **all** questions.
Write the answers in the spaces provided on the question paper.

SECTION B
Answer any **one** question.
Answer on the separate answer paper provided.
The answer should be illustrated by large, clearly labelled diagrams, where appropriate.

At the end of the examination:
Fasten all separate answer paper securely separate from the question paper.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
TOTAL	

Section A

Answer **all** questions.

QUESTION ONE

Fig. 1.1 represents the molecular structure of tubulin.

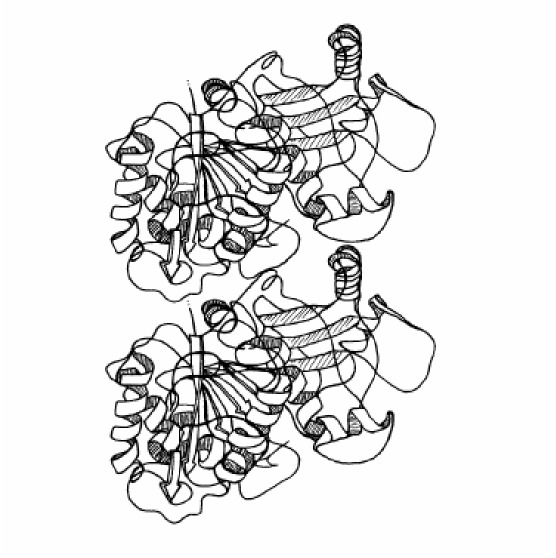


Fig. 1.1

(a) Describe the secondary, tertiary and quaternary structure of tubulin shown in Fig. 1.1.

____

____

____ [3]

(b) Tubulin is involved in the construction of microtubules.

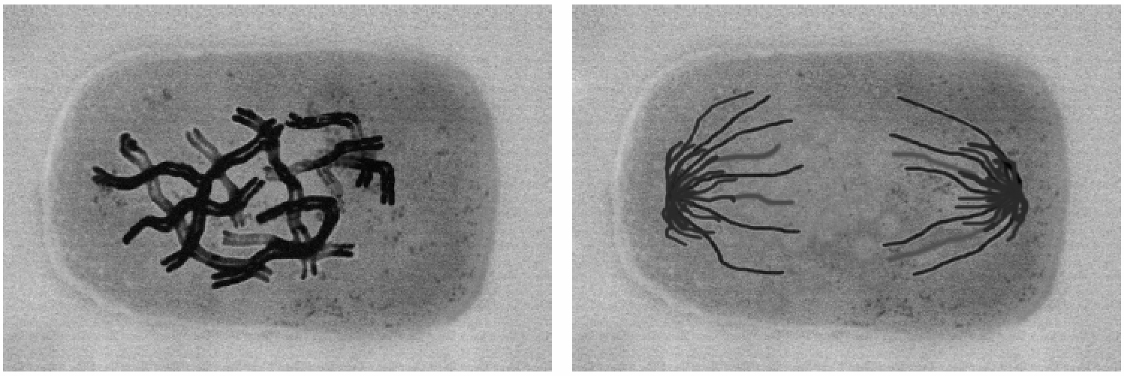
(i) Where in the cell do microtubules originate?

____ [1]

(ii) Describe the role of microtubules in M phase of the cell cycle.

____ [1]

Fig. 1.2 shows two stages of mitosis in a cell from a root tip of *Allium cepa.*



D E

Fig. 1.2

(c) Describe what happens to the chromosomes during mitosis between the stage shown in **D** and the stage shown in **E**.

____

____

____

____

____

____ [3]

(d) Describe the events that occur within a cell after the stage shown in **Fig. 1.2 E** to allow the formation of two cells.

____

____

____

____

____

____ [3]

(e) Explain how the events in the mitotic cell cycle ensure that all the cells in the root are genetically identical.

____

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____ [3]

[Total: 14]

QUESTION TWO

Fig. 2 shows events occurring during the synthesis of a protein that is secreted from a cell.

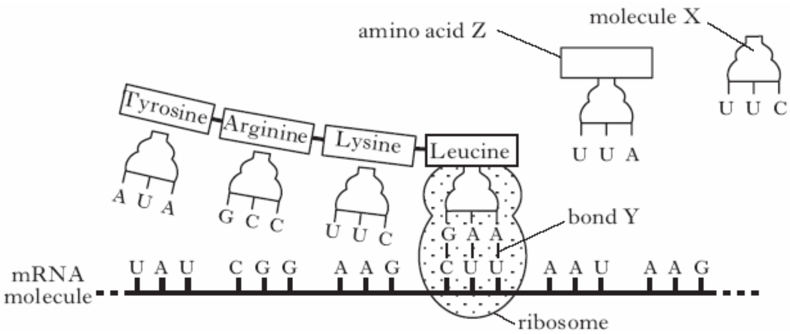


Fig. 2

(a) (i) Name molecule **X**. ______________________________________ [1]

(ii) Name bond **Y**. ____ [1]

(b) What name is given to a group of three bases on mRNA that codes for an amino acid?

____ [1]

(c) Give the sequence of DNA bases that codes for amino acid Z.

____ [1]

(d) Describe the roles of the endoplasmic reticulum and the Golgi apparatus between the synthesis of the protein and its release from the cell.

Endoplasmic reticulum ____

____ [1]

Golgi apparatus ____

____ [1]

(e) The table contains some information about the structure and function of proteins.
Add information to the boxes to complete the table.

Protein	Structure (Globular or Fibrous)	Function
Cellulase		
Collagen		Structural protein in skin

[3]
[Total: 9]

QUESTION THREE

Read carefully the passage about viruses and answer the questions that follow.

All viruses differ fundamentally from prokaryotes and eukaryotes and are considered to be nothing more than protein-coated nucleic acids.

All viruses are obligatory intracellular parasites: they enter the cells of their host and direct the synthesis of their own genetic material, which is either RNA or DNA, and viral-specific proteins. The latter assemble to form the protective capsid which surrounds the genetic material (genome) of the viral particles. These proteins are produced in the same way as normal host proteins.

Viral genomes can be single-stranded or double-stranded, linear or circular forms of either DNA or RNA. Retroviruses contain single-stranded RNA and the enzyme, reverse transcriptase. Once the virus is within its host, this enzyme synthesises DNA from the RNA template. The DNA then directs the synthesis of capsid proteins.

The genome of the tobacco mosaic virus (TMV) consists of a single strand of RNA with 6000 nucleotides. This codes for a protein; the helical capsid is composed of 2000 molecules of this protein. Bacterial viruses direct the synthesis of an enzyme, lysozyme, which permits the release of virus particles form the host bacterium. The capsid of the DNA-containing influenza virus is further surrounded by an envelope which consists of a lipid bilayer with glycoprotein molecules. This forms around the virus particles as they leave the host cell. The envelope enables the virus particles to gain entry to other cells in the host.

Some viruses, such as TMV, have no specific agent for transmission between hosts; they simply spread by leaf contact. Other plant viruses are transmitted by insects. Some are pollen transmitted and spread by cross-pollination from infected host plants. In this way they can infect both parent plants and the next generation.

(a) State **three** ways in which viruses differ structurally from prokaryotes.

1. ______________________________

2. ______________________________

3. ______________________________

[3]

(b) Explain why viruses are necessarily ***intracellular parasites***. (line 3)

____

____

____

______________________________ [3]

(c) (i) State the maximum number of amino acids that can be coded for by the genome of tobacco mosaic virus (TMV).

[1]

(ii) Explain your answer.

[2]

(d) (i) Give a reason for the synthesis of the enzyme lysozyme just prior to the release of virus particles from a bacterium.

[1]

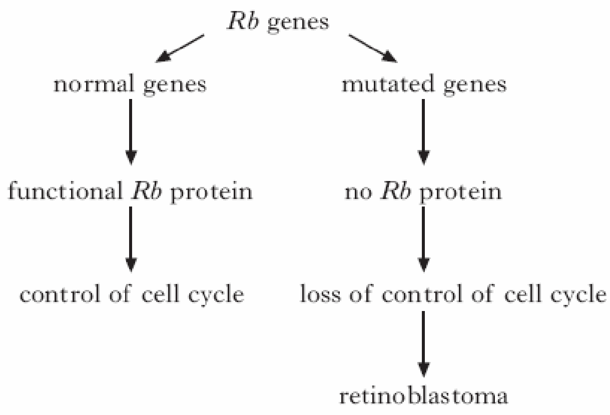
(ii) Suggest how the envelope which surrounds the influenza virus permits entry to host cells.

[2]

[Total: 12]

QUESTION FOUR

(a) *Retinoblastoma* is a rare cancer that develops in the eyes of children. Mutation in both copies of the retinoblastoma (*Rb*) gene results in proliferation of cells that would normally form retinal tissue. The protein arising from the *Rb* gene is abundant in the nucleus of all normal mammalian cells where it has an important role in the cell cycle.



When phosphorylated, the *Rb* protein binds to gene regulatory proteins and prevents them from activating cell proliferation. When unphosphorylated, it cannot bind to the gene regulatory proteins.

(i) Explain why kinase enzyme activity might restrict cell division when functional *Rb* protein is abundant.

____

____

____ [2]

(ii) Explain how the information provided indicates that the *Rb* gene is **not** a proto-oncogene.

____

____ [1]

(b) A transcription factor is a protein that promotes transcription at a particular locus. The gene *Ets 2* codes for a transcription factor, and is located on human chromosome 21. The gene is expressed in cartilage and bone tissues during development. In mice, *Ets 2* is located on chromosome 16 and mice with trisomy 16 show similar skeletal abnormalities to humans with Down's syndrome.

(i) Explain the genetic basis of Down's syndrome in humans.

____

____

____

______________________________ [3]

(ii) Explain what is meant by the term *trisomy;*

____

______________________________ [1]

(iii) Explain how a trisomy, such as trisomy 16 in mice, can occur in a zygote.

____

____

____

______________________________ [3]

In order to investigate the possible role of *Ets 2* in Down's syndrome, transgenic mice have been produced which carry an extra copy of mouse *Ets 2*. These mice show similar skeletal abnormalities to mice with trisomy 16. Tissues from the transgenic mice produced up to 1.8 times more *Ets 2* messenger RNA than control mice.

(iv) Suggest why these transgenic mice and mice with trisomy 16 show similar abnormalities.

____

____

____

______________________________ [2]

[Total: 12]

QUESTION FIVE

The coat colours of Labrador Retriever dogs are determined by two pairs of unlinked genes **B/b** and **E/e**. The genotype of black coated dogs is **B**—**E**—**,** brown **bbE**— and golden — — **ee** (where **—** indicates the presence of either the dominant or recessive allele).

(a) If heterozygous black (**BbEe**) dogs are mated, what phenotypes can be expected in the progeny and in what proportions?

Show how you derive your answer.

Expected phenotypes ____ [1]

Proportions ____ [1]

Derivation

[4]

The pedigree shown in **Fig. 3** below illustrates the inheritance of coat colour in Labrador Retrievers. Dogs with black coats are represented by solid symbols, those with brown coats by shaded symbols and those with golden coats by open symbols.

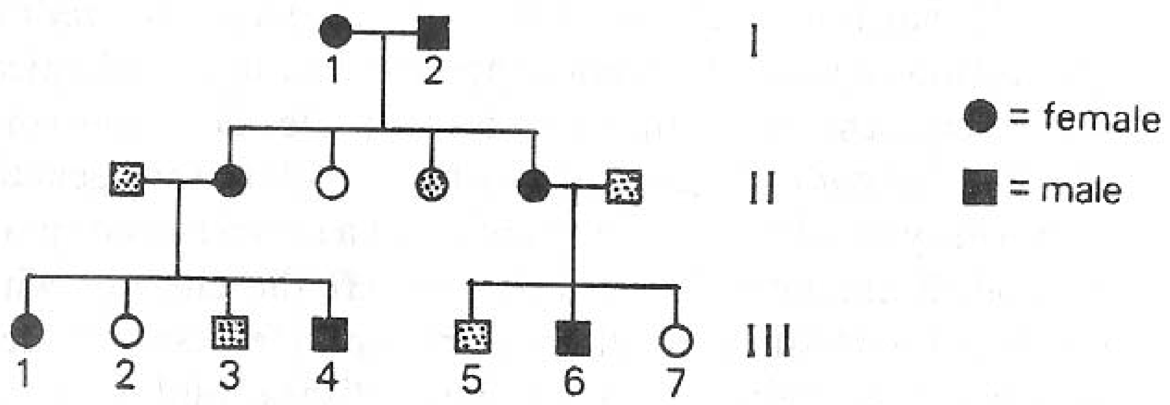


Fig. 3

(b) What is the most probable genotype of the individual III – 1 ?

Probable genotype ____________________ [1]

(c) What are the possible genotypes of III – 7 ?

Possible genotypes of III – 7 ____________________ [1]

Derivation ____________________

____

____________________ [2]

[Total: 10]

QUESTION SIX

Table 1 shows the rate of phosphate absorption by barley roots in a solution aerated with different mixtures of nitrogen and oxygen.

percentage of oxygen in aeration mixture	phosphate absorption/ $\mu mol\ g^{-1}\ h^{-1}$
0.1	0.07
0.3	0.15
0.9	0.27
2.1	0.32
21.0	0.33

Table 1

(a) State the conclusions that can be drawn about phosphate absorption from the data in **Table 1**.

____

____

____

____ [3]

Fig. 4 shows the rate of phosphate absorption by barley roots placed in solutions containing different concentrations of DNP (2,4-dinitrophenol). DNP is an uncoupler of the electron transport chain. Each solution was aerated with 21% oxygen.

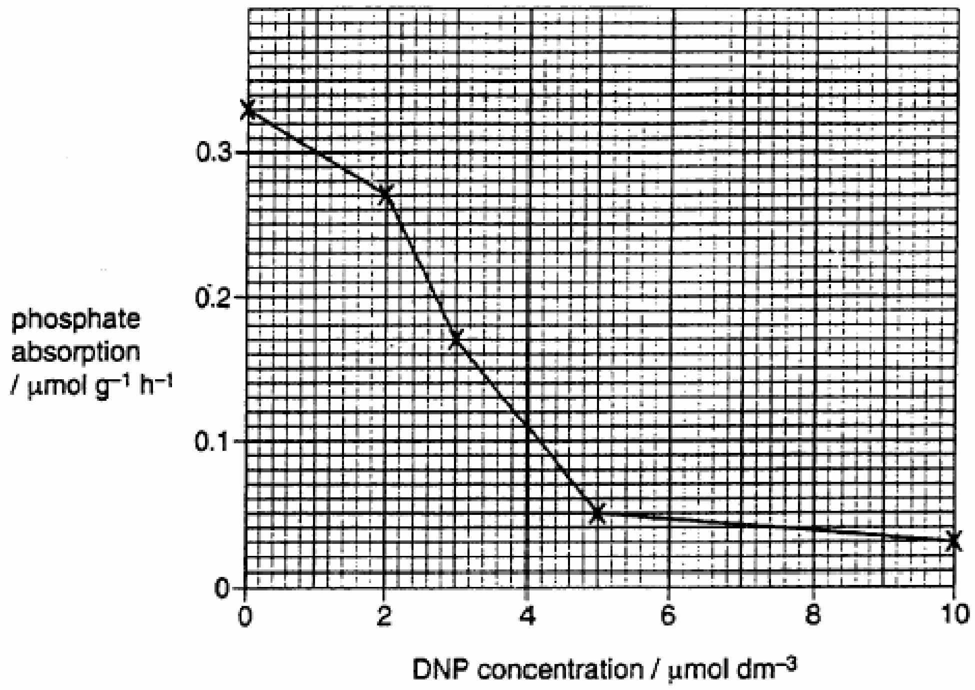


Fig. 4

(b) With reference to **Fig. 4**, describe and explain the effect on phosphate absorption of adding DNP to barley roots.

____

____

____

____

____ [4]

Malonate is an inhibitor of the Krebs cycle.

(c) (i) Predict the effect of adding malonate instead of DNP, on the uptake of phosphate by the barley roots.

[1]

(ii) Explain your answer to (i).

[3]

(d) State two uses of phosphate in living organisms.

[2]

[Total: 13]

QUESTION SEVEN

Sickle cell anaemia is a condition where the haemoglobin in the red blood cells can no longer carry oxygen efficiently at low oxygen concentrations. Heterozygous individuals show some symptoms of anaemia and appear to be at a potential disadvantage, but they have some resistance to malaria.

In **Fig. 5**, **A** shows the distribution of the malarial parasite in parts of East Africa below 1500 metres, whilst **B** shows the area in which more than 15% of the adults in the population are heterozygous for the sickle cell condition.

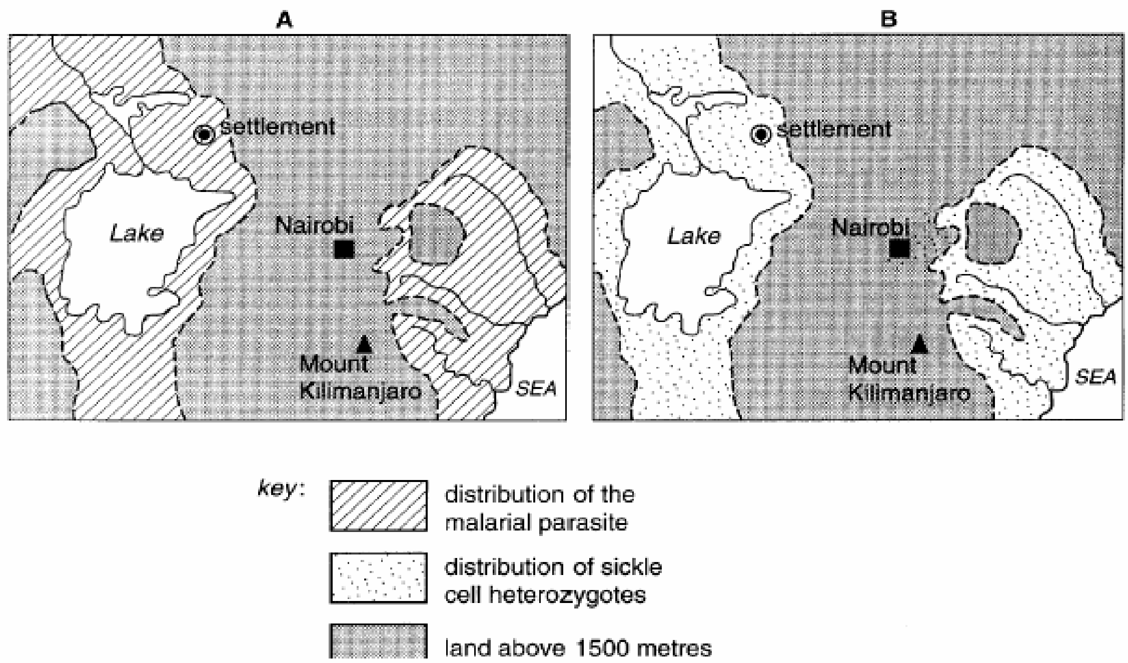


Fig. 5

(a) Explain briefly how the mutation which produces sickle cell anaemia may occur.

____

____

____

____ [3]

(b) Suggest why, and under what conditions, the red blood cells become sickle-shaped.

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____

____ [2]

(c) With reference to **Fig. 5**,

Describe, with reasons, the distribution of the sickle cell heterozygotes.

[2]

(d) If the inhabitants of the settlement indicated on **Fig. 5** were moved to an area where malaria did not exist, suggest what might happen to the population of heterozygotes in future generations. Explain your answer.

[3]

[Total: 10]

Section B

Answer **one** question.

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a), (b)** etc., as indicated in the question.

QUESTION EIGHT

(a) Describe the differences in structure between the prokaryotic and eukaryotic genome. [8]

(b) Describe the control mechanisms involved in the expression of the eukaryotic genes. [12]

QUESTION NINE

(a) Compare a molecule of triglyceride with a molecule of phospholipid. [5]

(b) Discuss the properties of a cell membrane that are important in the functioning of a mammalian nerve cell. [10]

(c) Suggest reasons why the endocrine system instead of the nervous system is used to control certain physiological processes. [5]